pgmuvi: Quick and easy Gaussian Process Regression for multi-wavelength astronomical timeseries

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Software

- Review
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Summary

Time-domain observations have long been central to astronomy, providing critical insights into various celestial phenomena, and their importance continues to grow as modern surveys expand the volume and scope of available data. They are often the only way to study certain objects. The volume of time-series data is increasing dramatically as new surveys come online - for example, the Vera Rubin Observatory will produce 15 terabytes of data per night, and its Legacy Survey of Space and Time (LSST) is expected to produce five-year lightcurves for $> 10^7$ sources, each consisting of 5 photometric bands. Historically, astronomers have worked with Fourier-based techniques such as the Lomb-Scargle periodogram or information-theoretic approaches; however, in recent years Bayesian and data-driven approaches such as Gaussian Process Regression (GPR) have gained traction. However, the computational complexity and steep learning curve of GPR has limited its adoption. `pgmuvi` makes GPR of multi-band timeseries accessible to astronomers by building on cutting-edge open-source machine-learning libraries, and hence `pgmuvi` retains the speed and flexibility of GPR while being easy to use. It provides easy access to GPU acceleration and scalable Gaussian Process modelling for multi-wavelength time series. While the current implementation focuses on efficient maximum a posteriori estimation, future versions will extend this framework to full Bayesian inference of the hyperparameters (e.g. the periods), enabling full posterior characterization of the variability.

Statement of need

Astronomical objects are in general not static, but vary in brightness over time. This is especially true for objects that are variable by nature, such as pulsating stars, or objects that are variable due to their orbital motion, such as eclipsing binaries. The study of these objects is called time-domain astronomy, and is a rapidly growing field. A wide range of approaches to time-series analysis have been developed, ranging from simple period-finding algorithms to more complex machine learning techniques (e.g. Donoso-Oliva et al., 2023; Friedman, 1984; Huijse et al., 2018; Palmer, 2009 and many more). Perhaps the most popular in astronomy is the Lomb-Scargle periodogram (Lomb, 1976; Scargle, 1982), which is a Fourier-based technique to find periodic signals in unevenly sampled data. However, the handling of unevenly sampled data is not the only challenge in time-series analysis.

A particular challenge in astronomy is handling heterogeneous, multiwavelength data (VanderPlas & Ivezić, 2015). Data must often be combined from a wide variety of instruments, telescopes or surveys, and so the systematics or noise properties of different datasets vary widely. In addition, by combining multiple wavelengths, we gain a better understanding of the physical processes driving the variability of the object. For example, some variability mechanisms differ as a function of wavelength only in amplitude (e.g. eclipsing binaries), while others may vary in phase (e.g. pulsating stars) or even period (e.g. multiperiodic systems). `pgmuvi` is particularly motivated by the need to interpret the lightcurves of long-period variables (such as AGB stars), where different radiation processes are relevant in different wavelength ranges, with changes in molecular bands dominating optical variability, while changes in dust radiation dominate the infrared. Their variations are caused by changes in stellar temperature and radius, but lead to changes in chemistry (changing the presence and depth of molecular absorption bands across the optical and near-infrared), the formation of circumstellar dust (changing the emission in the near- and mid-infrared) and changes in the ionisation state of circumstellar gas (causing changes in the chromospheric emission at ultraviolet and radio wavelengths). Each of these processes have different amplitudes and phase-lags compared to the changes in temperature and radius of the star itself, meaning that interpreting multiple wavelengths at once requires a model that can handle the differences across wavelengths. As a result, a complete physical picture can only be obtained by modelling all relevant wavelengths simultaneously.

Gaussian processes (GPs) have recently become a popular tool to handle these challenges. GPs are a flexible way to forward-model arbitrary signals, by assuming that the signal is drawn from a multivariate Gaussian distribution. By constructing a covariance function describing the covariance between any two points in the signal, we can model the signal as a Gaussian process. By doing so, we are freed from any assumptions about sampling, and can model the signal as a continuous function. We are also able to model the noise in the data, and thus account for heteroscedastic noise. We also gain the ability to directly handle multiple wavelengths, by constructing a multi-dimensional covariance function, allowing for simultaneous treatment of temporal and wavelength dependence. Hence, Gaussian Process Regression (GPR) is a machine learning technique that is able to model (quasi)-periodic signals in unevenly sampled data, and is thus well suited for the analysis of astronomical time-series data.

However, GPR is not without its challenges. The most popular covariance functions are often not tailored to modelling complex signals, which implies that users must construct their own covariance functions. GPs are also computationally expensive, and thus approximations must be used to scale to large datasets. Finally, many GP packages either have very steep learning curves or only provide limited features, and thus are not suitable for the average astronomer.

In this paper we present a new Python package, `pgmuvi`, to perform GPR on multi-wavelength astronomical time-series data. The package is designed to be easy to use, and provide a quick way to perform GPR on multi-wavelength data by exploiting uncommon but powerful kernels and appoximations. The package is designed to be flexible and allow the user to customize the GPR model to their needs. `pgmuvi` exploits multiple strategies to scale regression to large datasets, making it suitable for the current era of large-scale astronomical surveys.

State of the field

A number of other software packages exist to perform GPR, some of which, such as `celerite` (D. Foreman-Mackey et al., 2017; D. Foreman-Mackey, 2018), `tinygp` (Daniel Foreman-Mackey, 2023) or `george` (Ambikasaran et al., 2015) were developed within the astronomical community with astronomical time-series in mind. However, these each have their own limitations. `celerite` is extremely fast, but is limited to one-dimensional inputs, and thus cannot handle multiwavelength data, except under certain restrictive assumptions (all bands must have identical temporal sampling). Furthermore, since it achieves its speed through a specific form of kernel decomposition, it is not able to handle arbitrary covariance functions. It is therefore restricted to a small number of kernels with specific forms; while it is able to handle the most common astronomical timeseries by combining these kernels, not all signals can be modelled in this way. `tinygp` is a more general package, able to retain a high degree of flexibility while still being fast thanks to a JAX-based implementation, which makes it feasible to implement models in `tinygp` that are equivalent to those in `pgmuvi`. However, `pgmuvi` is designed to provide an easier learning curve by packaging GPs with data transforms and inference routines. In essence, `tinygp` could in principle be used by `pgmuvi` as a GP backend instead of GPyTorch. For a summary of the state of the art of GPR in astronomy, see the recent review by Aigrain & Foreman-Mackey (2023). A simple comparison between different GP packages showing some cases where `pgmuvi` is a good choice are included in the documentation, while Fig. Figure 1 and Table Figure 2 summarise some of that comparison.

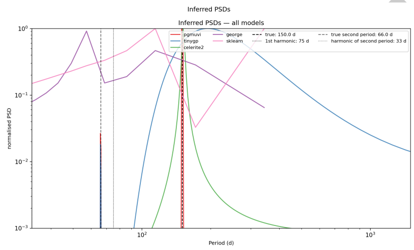

Figure 1: Comparison of the PSDs recovered by different GP packages for an arbitrary multiperiodic lightcuve. Not only does `pgmuvi` recover both periods accurately, it also localises with great precision.

Comparison of different GP packages.

Figure 2: Comparison of different GP packages.

Software design

`pgmuvi` builds on the popular GPyTorch library. GPyTorch (Gardner et al., 2018) is a Gaussian process library for PyTorch (Paszke et al., 2019), which is a popular machine learning library for Python. By default, `pgmuvi` exploits the highly-flexible Spectral Mixture kernel (Wilson & Adams, 2013) in GPyTorch, able to model a wide variety of signals. This kernel is particularly interesting for astronomical time-series data, as it can effectively model multi-periodic and quasi-periodic signals. The spectral mixture kernel models the power spectrum of the covariance matrix as Gaussian mixture model (GMM), making it highly flexible, easy to interpret and adaptable to multi-dimensional data. This kernel is known for its ability to extrapolate, and is thus well suited to cases where prediction is important (for example, preparing astronomical observations of variable stars). By modelling the power spectrum in this way, `pgmuvi` effectively filters out noise in the periodogram, and thus is able to find the dominant periods in noisy data more effectively than, for example, the Lomb-Scargle periodogram.

However, the flexibility of this kernel comes at a cost; for more than one component in the mixture, the solution space becomes highly non-convex, and the optimization of the kernel hyperparameters becomes difficult. `pgmuvi` addresses this by first exploiting the Lomb-Scargle periodogram to find the dominant periods in the data, and then using these periods as initial guesses for the means of the mixture components.

Multiple options are available to accelerate inference depending on the size of the dataset. For small datasets, the exact GPs can be used to scale to datasets of up to ~ 1000 points. `pgmuvi` can exploit the Structured Kernel Interpolation (SKI) approximation (Wilson & Nickisch, 2015) to scale to datasets of up to $\sim 10^5$ points. Future work will include implementing approximations for even larger datasets: `pgmuvi` can in principle exploit the Sparse Variational GP (SVGP) or Variational Nearest Neighbour (VNN) approximations (Hensman et al., 2013; Wu et al., 2022) to scale to datasets of arbitrary size. `pgmuvi` can employ also GPU computing for both exact and variational GPs.

Finally, by allowing arbitrary GPyTorch likelihoods to be used, `pgmuvi` can be extended to a wide range of problems. For example, an instance of `gpytorch.likelihoods.StudentTLikelihood` can be dropped in to turn `pgmuvi` into a T-Process regressor, or missing data can be handled using `gpytorch.likelihoods.GaussianLikelihoodWithMissingObs`.

To summarise, the key features of `pgmuvi` are:

- Highly flexible kernel, able to model a wide range of multiwavelength signals
- Able to exploit multiple strategies to scale to large datasets
- GPU acceleration for both exact and variational GPs
- Initialisation of kernel hyperparameters using Lomb-Scargle periodogram
- Automated creation of diagnostic and summary plots
- Automated reporting and visualization of fitted kernel hyperparameters

Research impact statement

`pgmuvi` is already in use for projects in our group. One of the authors' (DAVT) masters thesis served as the first test of the code and has analyzed thousands of light curves at optical and infrared wavelengths for over seven hundred dusty stars within 3 kpc of the Solar Neighborhood. The paper resulting from this work (Srinivasan et al., in prep) deals with the analysis of multiwavelength light curves for targets from the Nearby Evolved Stars Survey (NESS; Scicluna et al. (2022), https://evolvedstars.space).

AI usage disclosure

Github Copilot, Claude Code and ChatGPT were used in the development of `pgmuvi`. They were used to generate and review code under human instruction and review, write unit tests, and assist in writing and editing documentation.

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